PROJECT SYNOPSIS

Handwritten Digit Recognition Using Neural Networks

(MNIST Dataset)

1. Project Overview

The MNIST Digit Classification project is a beginner-friendly Artificial Intelligence and Machine Learning project that focuses on recognizing handwritten digits (0–9) using a Neural Network model. The system is trained on the MNIST dataset, which contains grayscale images of handwritten digits. The objective is to build an intelligent model capable of accurately identifying and classifying handwritten numbers. The project introduces students to image processing, deep learning, neural networks, model training, testing, and performance evaluation.

2. Project Objectives

- Understand the fundamentals of Neural Networks.
- Learn image classification techniques.
- Train a model using the MNIST dataset.
- Evaluate model accuracy and performance.
- Deploy a simple digit recognition system.

3. Dataset Information

Dataset: MNIST Handwritten Digits Dataset

Dataset characteristic	Description
Total images	70,000 digit images
Image dimensions	28 × 28 pixels
Image type	Grayscale
Digit classes	0 to 9
Training samples	60,000 images
Testing samples	10,000 images

4. Technologies Required

Programming Language: Python

Libraries

- NumPy
- Pandas
- Matplotlib
- TensorFlow
- Keras
- Scikit-learn

Development Environment: Jupyter Notebook, Google Colab, VS Code

5. Project Workflow

Phase 1: Data Collection

- Load the MNIST dataset.
- Explore image samples.

Phase 2: Data Preprocessing

- Normalize image values.
- Reshape image dimensions if required.

Phase 3: Model Development

- Design a Neural Network architecture.
- Configure input, hidden, and output layers.

Phase 4: Model Training

- Train the model using training data.
- Monitor accuracy and loss.

Phase 5: Testing and Evaluation

- Test the model on unseen images.
- Generate performance metrics.

Phase 6: Prediction

- Allow users to input handwritten digits.
- Display predicted output.

6. Expected Outcomes

- Digit recognition accuracy above 95%.
- Practical understanding of image classification.
- Experience working with deep learning frameworks.

7. Applications

- Bank cheque processing.
- Postal code recognition.
- OCR systems.
- Educational tools.
- Automated form processing.

8. Project Deliverables

The project deliverables specified in the manual are:

Documentation

- Project Report (Word/PDF)
- Synopsis
- Abstract

Presentation

- PowerPoint Presentation (PPT)

Source Code

- Python Files (.py)
- Jupyter Notebook (.ipynb)

Screenshots

- Training Results
- Accuracy Graphs
- Prediction Outputs

Prepared according to Project 1 in the AI Project Guide Manual.